

Feedback and Newcomer Retention in X Community Notes

CS 5306 / INFO 5306, Project 1

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Abstract

In X's Community Notes, a contributor's first note can be rated Helpful, rated Not Helpful, or left in "Needs More Ratings" (NMR) with no verdict. We ask whether a Not Helpful verdict on a first note is associated with different retention than no verdict. Studies of Wikipedia newcomers cannot make this comparison, because Wikipedia has no equivalent of a contribution that never receives a verdict. Using a public snapshot of 3.1 million notes, we classify each author's first note by its status seven days after it was written. Authors whose first note had been rated Not Helpful by then were 15.0 percentage points less likely to write another note within 30 days than authors whose note was still in NMR (95% CI -15.5 to -14.5). Most of this difference coincides with a platform rule. Since April 2024, a single Not Helpful note removes a new author's ability to write, and 89.4% of authors whose first note was rated Not Helpful have lost writing ability, usually on the day the verdict arrived. The difference is 6.1 points for cohorts that started before the rule and 17.1 points for cohorts that started after it. In the pre-rule cohorts, at least 4.9 points of the difference cannot be explained by the lockout.

1 Introduction

Crowdsourced moderation depends on contributors who keep participating after their first attempt. In Community Notes, volunteers write short notes that add context to posts on X, and other volunteers rate those notes. A matrix-factorization model publishes a note only when raters who usually disagree with each other both find it helpful [4]. The bar is high, and most notes never clear it.

As a result, a first note has three possible outcomes: it is rated Helpful, it is rated Not Helpful, or it receives no verdict and stays in Needs More Ratings (NMR) indefinitely. Research on Wikipedia has shown that negative early feedback predicts newcomer departure: newcomers whose first edits are reverted are less likely to keep editing [1, 2]. On Wikipedia, however, an edit that is not reverted stays live in the article, so escaping rejection amounts to acceptance; there is no state in which a contribution is neither accepted nor rejected. In Community Notes, that unresolved state is the most common outcome.

Our question is whether an explicit rejection (Not Helpful) is associated with different retention than no verdict (NMR). Helpful serves as a reference point. The question matters for design. A system in which most contributions never receive a verdict has in effect chosen a default response to newcomers, and it is not obvious whether that response is better or worse for retention than a rejection.

We find that a Not Helpful first note predicts much lower retention than an NMR first note. A large part of this difference comes from a platform rule we did not know about when we began: since

April 2024, a Not Helpful first note automatically removes the author's ability to write notes. Section 6.2 separates the part of the difference that this rule accounts for from the rest. In cohorts that started before the rule, a difference of about five points remains.

2 Background

Two platform rules matter for interpreting our results.

Contributors must rate before they can write. New contributors can only rate notes. They gain the ability to write when their Rating Impact reaches 5. Rating Impact increases when a contributor rates a note before it receives a status and the rating matches the status the note later receives [6]. This requirement was introduced in September 2022 and applies to every cohort in our analysis. The authors we call newcomers are therefore new to writing only. Each of them has already rated enough notes, early and accurately enough, to pass this threshold. They are probably more committed than a typical first-time Wikipedia editor, which weakens the comparison to that literature.

A Not Helpful verdict can remove the ability to write. A contributor loses writing ability when three of their last five resolved notes are rated Not Helpful. Since April 2, 2024, they also lose it when any of their notes is rated Not Helpful while their Writing Impact is zero or negative [6]. Writing Impact starts at zero and changes only when a note resolves, so a first-time author's Writing Impact is exactly zero. From April 2024 onward, a single Not Helpful first note is therefore enough to remove writing ability, while an NMR first note has no consequence. In other words, the platform itself treats the two outcomes we compare differently. We return to this rule in Section 6.2.

Statuses are recomputed periodically and can change, but the public data records when each note first left NMR and which status it received [7]. We use that first transition.

3 Related Work

The most relevant prior work studies how Wikipedia retains its newcomers. Halfaker et al. [2] analyzed 400,000 revisions and found that having early edits reverted strongly reduces how much newcomers contribute afterward, although reverting also improves the overall quality of the encyclopedia. A later study [1] attributes much of Wikipedia's decline to the increasingly automated and impersonal rejection of newcomers' edits.

Community Notes differs from Wikipedia in two ways that matter for our question. First, rejection comes from a scoring algorithm rather than from another editor, so a rejected author has no one to respond to and no explanation to read. Second, the most common outcome is no verdict at all, which Wikipedia does not produce at a comparable scale. Mohammadi et al. [3] describe the program's first four years, including contributor activity and rating behavior.

To our knowledge, no prior work has examined how the absence of a verdict relates to retention.

4 Data

We use the public Community Notes data downloads [5] from the snapshot released on **September 18, 2026**. Each snapshot covers all notes back to 2021, so one is enough. It includes only notes created at least 48 hours before release, so our data ends on **September 16, 2026**, and all censoring uses that cutoff.

notes (3,086,867 rows) records each note’s author and creation time. noteStatusHistory records its outcome in two fields: firstNonNMRStatus, the first status other than NMR, and timestampMillisOfFirstNonNMRStatus, when it arrived. Both are empty for notes that never left NMR, which is the case for 82.7% of notes. Deleted notes disappear from notes but keep their status history, so our authors are drawn only from notes that were not deleted. userEnrollment records each contributor’s current enrollment state, which we use to identify lost writing ability and automated accounts. We did not use the ratings table, which records individual ratings rather than note outcomes, or the text of the annotated posts, which is not public.

Author identifiers are stable. Our analysis assumes an author identifier refers to the same person in every release. Snapshots expire after about a week, so this can be checked only while two are available. Of 3,282,773 notes present in both the September 18 and September 16 releases, none changed author.

Two gaps in the outcome data. First, firstNonNMRStatus is empty for every note created before May 2022 (28,988 notes), so those cohorts look 100% NMR only because no verdicts were recorded. Second, 233,514 notes (7.6%) have no noteStatusHistory row at all. Their outcome is unknown, which is not the same as never leaving NMR. They appear from March 2024 onward, and over those 28 months their share of each month’s notes rises from 6.7% to 14.1%, so counting them as NMR would create a spurious trend across cohorts. Both gaps lead to exclusions (Section 5.4).

5 Method

5.1 Contributor table

We join notes to noteStatusHistory on note ID and build one row per author: the first note’s outcome and when it arrived, when the author next wrote, their cohort (the month of the first note), and features of the first note (classification, URL presence, length, and reason tags). The first note is the author’s earliest in the snapshot, with ties broken by note ID. Because the snapshot reaches back to 2021, it is the author’s actual first note. The table has 343,176 rows.

The table has exactly one row per author, its note counts sum to the number of rows in notes, and no verdict or second note is dated before a first note. Checks written against the query that built the table could share its bugs, so we also rebuilt the rows of twenty randomly chosen authors from the raw files with a separate Python parser. Every field of every row matched.

5.2 Landmark classification

We classify each first note by **its status seven days after it was written**, not by the status it eventually received. An eventual status can reflect events that had not happened when the author decided

Table 1: Retention by first-note outcome at day 7. Wilson 95% intervals.

First-note outcome	<i>n</i>	30-day	90-day
Helpful	24,798	27.6%	43.7%
Not Helpful	19,153	10.6%	19.3%
No verdict (NMR)	183,373	25.6%	41.4%

whether to write again: a note that sat in NMR for three months before being rated Not Helpful gave its author no verdict while they were deciding. Whether a note is still in NMR at the end of the data also depends on when the snapshot was taken.

Seven days is long enough for nearly all verdicts. For first notes written in 2023 or later, the median wait is under eight hours and the 90th percentile is under two days. Verdicts that arrive after day seven count as NMR. Authors who wrote a second note before day seven are excluded, because they returned before the status we classify existed.

5.3 Retention and censoring

An author is retained if they write another note between day seven and the end of a 30-, 60-, or 90-day window. Each window includes only authors whose window closes before the data cutoff, so sample sizes differ across windows. Proportions have Wilson 95% intervals and differences have Newcombe hybrid-score intervals.

5.4 Exclusions

Of 343,176 authors, 322,094 remain after three exclusions:

- 5,052 authors whose first note was written before January 2023, when verdicts are either missing or were changed by X’s rescoring of all historical notes.
- 15,997 authors whose first note has no status-history row.
- 35 AI note-writer accounts, identified by an enrollment state beginning with api. They wrote 8.6% of all notes, and an automated account that stops writing says nothing about people leaving. Enrollment state is current rather than historical, so an AI account that has since changed state would be missed.

The day-7 landmark then leaves 230,588 authors, of whom 227,324 have a complete 30-day window.

6 Results

Authors whose first note was rated Not Helpful within seven days were 15.0 percentage points less likely than authors whose first note was still in NMR to write another note in the following 30 days (95% CI -15.5 to -14.5 ; $n = 19,153$ Not Helpful and 183,373 NMR). Section 6.2 shows that most of this difference coincides with the writing lockout.

6.1 Not Helpful versus no verdict

Table 1 and Figure 1 show that the two outcomes differ from NMR by very different amounts. A Not Helpful first note is associated with much lower retention than an NMR first note: 15.0 points lower at 30 days and 22.1 points lower at 90 days. A Helpful first

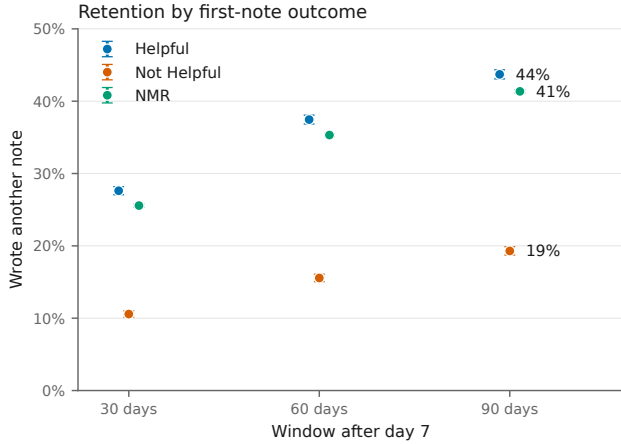


Figure 1: Retention by first-note outcome at 30, 60, and 90 days, with 95% intervals.

note is associated with only slightly higher retention than an NMR first note: 2.1 points higher at 30 days and 2.4 points higher at 90 days. Authors who received no verdict returned at nearly the same rate as authors whose notes were rated Helpful.

6.2 Cohorts and the April 2024 lockdown rule

Figure 2 shows the difference by cohort month. Through 2023 and early 2024, the difference is small and often not distinguishable from zero. It then widens sharply. Thirty-day retention for Not Helpful authors falls from about 28% in February 2024 to 9.6% in April 2024 and stays between 3% and 8% afterward, while retention for the other two groups declines together and much more gradually. Pooling the cohorts on each side of April 2024, the Not Helpful minus NMR difference is -6.1 points before (95% CI -7.4 to -4.7) and -17.1 points after (-17.6 to -16.7).

April 2024 is when X began removing writing ability after a single Not Helpful note (Section 2). The enrollment data shows the rule in operation. Of authors whose first note was rated Not Helpful, 89.4% have lost writing ability at least once, compared with 16.8% of authors whose first note stayed in NMR, and 48.7% are still unable to write. Among Not Helpful authors who wrote only one note and later lost writing ability, the median time between the verdict and the loss of writing ability is 0.03 days (under an hour), and 84.0% lost it within a week. For most of these authors, the verdict and the loss of writing ability happened at essentially the same time.

The lockdown does not explain the whole difference. The enrollment table records only a contributor’s most recent loss of writing ability, so an author who started in 2023 may have lost it years later under the new rule. Counting only losses recorded before April 2, 2024, 1.2% of pre-rule Not Helpful authors had lost writing ability during the period we study, compared with 0.3% of pre-rule NMR authors. Nearly all pre-rule authors were still able to write, and their difference is still 6.1 points. Even if every author who lost writing ability would otherwise have returned, the lockdown can

Table 2: Not Helpful minus NMR difference at 30 days under each variation.

Variation	Difference
Baseline (day-7 landmark)	-15.0 pp
Landmark at 3 days	-18.2 pp
Landmark at 14 days	-11.7 pp
Excluding not-misleading first notes	-14.0 pp
Retention = two or more later notes	-5.7 pp
Cohorts with 90+ days after the window	-15.0 pp
Eventual-status specification	-21.1 pp

account for at most 1.2 points of that difference, which leaves at least 4.9 points that it does not explain.

We therefore report two estimates that answer different questions. The pooled difference of 15.0 points describes what happens today to newcomers whose first note is rated Not Helpful, and most of it reflects the lockdown rule. The difference of 4.9 points in pre-rule cohorts is our estimate of the association between rejection and not returning when the lockdown does not apply.

6.3 Robustness

Table 2 repeats the main comparison under five alternative specifications. The Not Helpful minus NMR difference is negative in every case, ranging from -5.7 to -18.2 points. Moving the landmark shifts it in the expected direction: an earlier landmark keeps more authors but counts as NMR some notes that later receive a verdict, and a later landmark does the reverse. Requiring two or more later notes shrinks the difference because it lowers retention in every group. None of these shifts is as large as the gap between the pre-rule and post-rule cohorts.

Skipping the landmark inflates the difference to 21.1 points. That specification labels each note with its eventual status and measures retention from the day the note was written, which brings back the authors who wrote again within the first week. They are the most active authors in the data, and they returned before any verdict existed. The censoring rules are the same, so the gap comes from the landmark alone.

Kaplan–Meier estimates confirm that dropping incomplete windows does not drive the results. They start follow-up at day seven and censor authors at the data cutoff. By day 90, they show 43.7% of Helpful, 41.2% of NMR, and 19.0% of Not Helpful authors writing again, each within half a point of Table 1.

Controlling for observable features of the note does not shrink the difference. A logistic regression of 30-day retention on day-7 status, with cohort-month fixed effects and controls for the first note’s classification, URL presence, length, and reason tags, gives an odds ratio for Not Helpful of 0.36 (95% CI 0.34–0.38), an average marginal change of -18.5 points, 3.5 points larger than the unadjusted difference. Among the controls, a first note stating that a post is *not* misleading is associated with 2.4 points higher retention at the same status, even though such notes cannot be rated Helpful and are rated Not Helpful much more often.

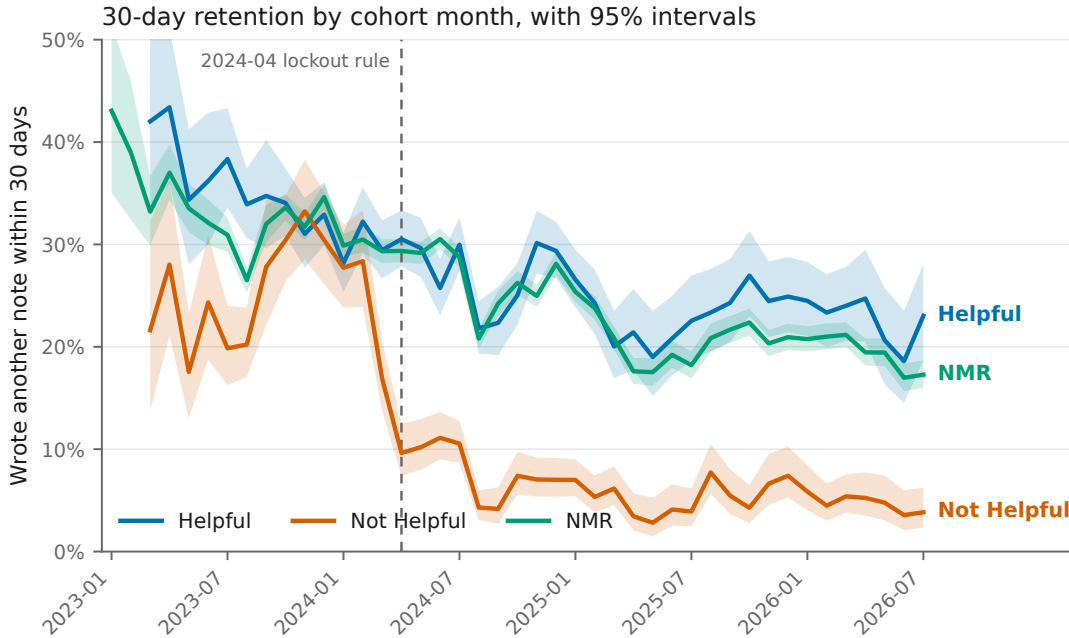


Figure 2: 30-day retention by cohort month and first-note outcome, with 95% intervals. The dashed line marks April 2024, when a single Not Helpful note began to remove writing ability. Cohort months too recent for a 30-day window to close before the data cutoff are omitted.

7 Limitations

Selection. We did not assign authors to outcomes. A weak first note may both be rated Not Helpful and come from an author who was leaving anyway, and controlling for observable features of the note does not rule this out. Both the pooled and the pre-rule differences are associations, not effects of rejection.

The lockout after April 2024. Section 6.2 bounds the lockout’s contribution only before the rule. After it, rejection and lockout cannot be separated, because userEnrollment records each contributor’s current state with no history. An author who was locked out when they would have returned looks the same as one who had already decided to stop, and one who lost writing ability, regained it, and then left looks the same as one who never lost it. This is why the smaller, older pre-rule sample carries our estimate of rejection without the lockout.

Whether authors saw the verdict. The landmark assumes that seven days is enough for an author to learn their note’s status. That depends on X’s notifications, which public data does not record. An author who never saw a Not Helpful verdict experienced it as no verdict, which would make the two groups look more alike than they were to the authors.

8 Conclusion

A Not Helpful verdict on a first note predicts much lower continued contribution than no verdict: 15.0 points lower at 30 days. Most of this difference coincides with X’s writing lockout. In cohorts that started before the lockout rule, when a rejection did not remove the ability to write, the difference is about five points. Rejection

on its own is associated with a modest drop in retention, and the lockout accounts for most of the rest of the pooled difference.

We did not expect how little a Helpful verdict adds. Authors whose first note was rated Helpful returned only about two points more often than authors who received no verdict. This has two implications for a system in which 82.7% of notes never receive a verdict. Receiving no verdict is not associated with the drop in retention that follows a rejection, so the large share of unresolved notes does not appear to be costing the system contributors. But a Helpful verdict is also associated with little additional retention. Our results therefore give little reason to expect faster verdicts to retain more new authors. A more relevant design question is whether a first rejection should remove the ability to write at all, given that 89.4% of authors whose first note is rated Not Helpful lose that ability, usually on the same day.

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A What We Did

Most of our four weeks went into understanding the system and the data rather than into the statistical analysis.

In the first week we worked out how the system operates. Reading the scoring source code alongside the published guide is how we found the April 2024 lockout rule and the date it took effect. Knowing the date is what made the before-and-after comparison in Section 6.2 possible; without it, we could only have listed the lockout as a confound. The guide’s web pages are rendered client-side and return no content to a plain HTTP request, so we worked from the Markdown files the site is built from.

In the second week we built the contributor table and found the two data problems described in Section 4. Verdict fields are empty before May 2022, which ruled out the 2021 and 2022 cohorts. And 7.6% of notes have no status-history row, which is not documented and would have added a growing number of unknown outcomes to the NMR group. We stopped to investigate each problem before continuing, which took about a day each. The same week, we checked that author identifiers were stable across two releases, which is only possible while both releases can still be downloaded.

Several infrastructure problems also took time. Our Python environment is a pinned Nix flake. Our first attempt used `uv`, which failed because its downloaded Python interpreter is dynamically linked against a loader that does not exist on NixOS. The survival-analysis library we planned to use did not build with the pinned packages, so we computed the Kaplan–Meier estimates and the log-rank test with `statsmodels`. Reading the tables with default CSV quoting silently corrupted rows, because some note text contains unescaped double quotes.

We also caught two of our own mistakes. The first version of the cohort figure showed a spike in every series in the most recent month. Only authors who joined in the first few days of that month had a 30-day window that closed before the cutoff, so that month’s estimate came from a biased subset of authors. We fixed this in the analysis with an explicit completeness flag rather than by trimming the chart. Second, our first draft of the regression results said the adjusted and unadjusted estimates agreed, when they differ by 3.5 points.

We kept the pipeline reproducible throughout. The contributor table was frozen after validation and never rebuilt, all exclusions are applied by one function, and the numbers in this report come from files generated by the analysis scripts rather than from manual calculation.

B Use of AI

We used Claude Opus 5 in Claude Code at every stage of the project, under our direction and review. It located and downloaded the data snapshot, which we checked against the published data dictionary

and a second release. It read the scoring source code and documentation to summarize the platform rules, including the lockout rule, and wrote the Python and DuckDB code that ingests the data, builds the contributor table, and runs the analyses. It also drafted the report, which we edited, and Claude Opus 5.5 helped revise it.

The decisions were ours. Work stopped at three review points: after the platform rules were summarized, when we decided which cohorts and accounts to exclude; after the contributor table was validated, when we accepted and froze it; and after the analysis, when we chose how to present the results. The report’s numbers come from the analysis scripts’ output, not from text the model wrote.